

EchoSign AI — Live Stage Demo & Sign Guide

Failsafe Hand Gestures, Sentence Reconstruction & Presentation Playbook

Bootcamp on Artificial Intelligence by NIELIT Delhi | Stage Ready Handout

1. The 3 Stage Golden Rules (Zero Failure Guarantee)

1. Camera Distance	Sit 2.5 to 3 feet away from your webcam. Ensure your Face and Chest are both clearly framed on screen so body anchors (chest, lips) trigger accurately.
2. Room Lighting	Ensure ambient light falls on the front of your face and hands. Avoid strong backlight (e.g. sitting with a bright window behind you).
3. Hold for 1–2 Seconds	EchoSign uses a deque(maxlen=7) consensus filter to eliminate camera jitter. Hold each gesture steady for 1.5 seconds until the HUD locks and speaks!

2. Complete Catalog of Failsafe Signs & Spoken Sentences

Category A: Healthcare & Medical Urgency (Hospital / Pain / Doctor)

Gesture / Combo	Physical Hand Movement	AI Spoken Output Sentence	Emotion & Urgency
PAIN + DOCTOR (Emergency)	1. Point index finger to chest/stomach (PAIN). 2. Tap 2 fingers on opposite wrist pulse (DOCTOR).	'I am experiencing pain and would like to see a medical professional.'	Medical Attention Urgency: 0.50 (Moderate)
PLEASE + DOCTOR (Care Request)	1. Flat open hand across chest (PLEASE). 2. Tap opposite wrist or neck pulse (DOCTOR).	'Could you please assist me in seeing a doctor?'	Care Requested Urgency: 0.40
DOCTOR (Single Sign)	Index & Middle fingers tap opposite wrist pulse or neck pulse.	'Could you please connect me with a doctor or medical professional?'	Medical Attention Urgency: 0.40
PAIN (Single Sign)	Index finger points directly towards chest/stomach area.	'I am experiencing some physical discomfort or pain.'	Care Requested Urgency: 0.40

Category B: Basic Needs (Food, Water, Hunger)

Gesture / Combo	Physical Hand Movement	AI Spoken Output Sentence	Emotion & Urgency
PLEASE + WATER (Drink Request)	1. Flat hand on chest (PLEASE). 2. 'W' shape (3 fingers up) touches lips/chin (WATER).	'Could I please get some drinking water?'	Polite Request Urgency: 0.20
PLEASE + FOOD (Meal Request)	1. Flat hand on chest (PLEASE). 2. Pinch 5 fingertips together touching lips/mouth (FOOD).	'Could I please have some food?'	Polite Request Urgency: 0.20
WATER + FOOD (Complete Need)	1. 'W' shape at lips (WATER). 2. Pinch cluster touching lips (FOOD).	'Could I please get some drinking water and food?'	Polite Request Urgency: 0.20
FOOD (Single Sign)	Pinch all 5 fingertips together and touch lips/mouth repeatedly.	'I am feeling hungry, could you please provide some food?'	Request / Need Urgency: 0.20
WATER (Single Sign)	'W' shape (Index, Middle, Ring up) tapped near chin/lips.	'Could I please have a glass of drinking water?'	Request / Need Urgency: 0.20

Category C: Greetings & Politeness (Respect & Gratitude)

Gesture / Combo	Physical Hand Movement	AI Spoken Output Sentence	Emotion & Urgency
NAMASTE (Dual Hand)	Bring both palms together at chest level with fingers pointing upward.	'Namaste! Welcome and heartfelt greetings to you.'	Warm & Grateful Confidence: 0.98
HELLO (Single Hand)	Open 5 fingers upright facing camera, wave near temple height.	'Hello! Wishing you a peaceful and pleasant day.'	Conversational Confidence: 0.97
NAMASTE + PLEASE	1. Both palms together (NAMASTE). 2. Flat open palm on chest (PLEASE).	'Namaste! Could you please assist me for a moment?'	Polite Greeting
NAMASTE + THANK YOU	1. Both palms together (NAMASTE). 2. Flat hand moving outward from lips (THANK YOU).	'Namaste! Thank you very much for your time and kind support.'	Warm & Grateful
THANK YOU (Single Sign)	Fingers of flat hand touch lips/chin and extend outward towards camera.	'Thank you so much! Truly appreciate your kind support.'	Warm & Grateful

Category D: Confirmation, Social & Agreement (OK / Yes / Love / Peace)

Gesture / Combo	Physical Hand Movement	AI Spoken Output Sentence	Emotion & Urgency
OK / GOOD (■ Sign)	Thumb and Index finger tips touch forming a circle; Middle, Ring, Pinky upright.	'All is well, everything is completely okay.'	Positive Confidence: 0.96
YES (Thumbs Up ■)	Closed fist with thumb upright pointing to the ceiling.	'Yes, absolutely correct.'	Agreement Confidence: 0.95
YES + THANK YOU	1. Thumbs Up (YES). 2. Flat hand extending from lips (THANK YOU).	'Yes, that is correct! Thank you so much.'	Warm & Grateful
NO + THANK YOU	1. Index & Middle finger snap down (NO). 2. Flat hand lips outward (THANK YOU).	'No, thank you, but I appreciate your offer.'	Polite Refusal
I LOVE YOU (■ Sign)	Thumb, Index, and Pinky fingers open simultaneously (Middle & Ring curled).	'I love and appreciate you deeply!'	Social / Affection Confidence: 0.96
PEACE (♌ V-Sign)	Index and Middle finger spread wide in 'V' shape (Ring, Pinky, Thumb tucked).	'Peace, calmness, and goodwill to all!'	Calmness Confidence: 0.96

Category E: ASL Fingerspelling & Dynamic Word/Name Pronunciation

When in **ASL Alphabet Mode (RTX 4060 GPU)**, fingerspelling individual letters triggers real-time GPU inference + geometric verification, automatically assembled into spoken words or custom names by our NLP Engine:

Letter	ASL Hand Pose Description	Key Finger Landmark Configuration
A	Make a closed fist with knuckles facing front; thumb stands upright beside the index finger.	All 4 fingers curled (ext=False), Thumb upright beside index (pts[4].y < pts[3].y).
Y	The classic 'Hang Loose' / 'Shaka' sign. Thumb & Pinky extended outward; middle 3 fingers curled.	Pinky & Thumb extended outward (ext_pinky & ext_thumb), Index, Middle, Ring curled.
U	Index and Middle fingers pointing straight UP and touching together (glued together); Ring & Pinky tucked.	Index & Middle extended upright with tip distance d < 0.20; Ring & Pinky curled.
S	Tight closed fist with thumb folded horizontally across the front of the curled fingers (like a punch).	All 4 fingers curled into fist, Thumb folded across the front of the fingers.
H	Index and Middle fingers extended horizontally to the side (pointing sideways); Ring & Pinky curled.	Index & Middle pointing sideways (dx > dy); Ring & Pinky curled.
B	All 4 fingers (Index, Middle, Ring, Pinky) upright together; thumb folded across the palm.	All 4 fingers extended upright together; Thumb tucked across the palm.
I	Only Pinky finger pointing straight UP; all other 4 fingers curled into a fist.	Pinky extended straight UP; Index, Middle, Ring, Thumb curled into fist.

Dynamic NLP Fingerspelling Assembly: Individual letters are continuously streamed and assembled into complete words or custom names.

■ **Text-to-Speech Output:** The voice smoothly pronounces the assembled word/name as complete, natural speech (e.g., custom names, technical terms, or unlisted vocabulary).

Control Tokens Supported: SPACE (open flat hand to separate words) • DEL (backspace to remove last letter).

3. Reverse Channel (Voice → Sign) Demonstration Phrases

To showcase Channel B (Hearing to Deaf), switch to the **'Voice → Sign'** tab, click the microphone, and speak any of these realistic phrases. The NLP tokenizer isolates keywords and pops up animated visual Sign Cards:

Scenario	Spoken English Line (Speak into Mic)	Visual Sign Cards Displayed
Hospital Urgency	'Please call a doctor, I have pain.'	Triggers 3 Sign Cards: PLEASE (Heart), DOCTOR (Cross), PAIN (Activity)
Meal / Hospitality	'Are you hungry? Do you need food and water?'	Triggers 2 Sign Cards: FOOD (Utensils), WATER (Droplet)
Social Respect	'Namaste, thank you for coming today.'	Triggers 2 Sign Cards: NAMASTE (Pranam), THANK YOU (Handshake)
Polite Check	'Hello, is everything okay with you?'	Triggers 2 Sign Cards: HELLO (Wave), OK (Checkmark)

4. Minute-by-Minute Stage Presentation Script

Follow this exact script when presenting before evaluators to deliver a captivating 4-minute presentation:

Time	Section Phase	Spoken Stage Lines & Live Actions
00:00 - 00:45	Introduction & Problem Statement	'Good morning respected evaluators. Over 466 million people face disabling hearing and speech impairments. Most academic sign projects only recognize static alphabet letters one-way. Today, we present EchoSign AI—a real-time, two-way assistive communication bridge powered by Computer Vision, NLP, and NVIDIA RTX 4060 GPU Deep Learning.'
00:45 - 01:45	Channel A: Dual-Hand Namaste & Daily Needs Demo	'First, I demonstrate Channel A: Deaf to Hearing. I bring both hands together in front of the chest. Notice the dual-hand tracking highlighting primary hand in Neon Cyan and secondary in Emerald, speaking Namaste! Next, I place a flat hand on my chest for PLEASE, followed by W-shape at my lips for WATER. Our NLP engine seamlessly reconstructs: 'Could I please get some drinking water?' with polite request sentiment.'
01:45 - 02:45	Medical Triage Intent & Social Confirmation	'In emergency medical scenarios, pointing to pain and tapping my wrist pulse reconstructs: 'I am experiencing pain and would like to see a medical professional' and tags a moderate urgency score. For confirmations, making the OK circle sign or I Love You gesture is classified with over 96% confidence.'
02:45 - 03:30	Channel B: Voice to Sign & RTX 4060 GPU Showcase	'Now, I demonstrate our reverse channel for hearing individuals. I speak into the mic: 'Please call a doctor, I have pain.' Instantly, animated visual Sign Cards appear for the deaf person. Finally, in ASL Alphabet Mode, our custom ASLResNet neural network classifies fingerspelled letters L and V directly on the RTX 4060 GPU in under 2 milliseconds with 99.87% accuracy.'
03:30 - 04:00	Conclusion & Q&A; Invitation	'All conversation logs are stored in our SQLite database for health analytics. EchoSign AI bridges human communication through accessible edge AI. Thank you, we are now ready for your questions.'